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K J SOMAIYA INSTITUTE OF TECHNOLOGY, SION, MUMBAI

DEPARTMENT OF COMPUTER ENGINEERING

AgriSense: AI-Driven Crop Monitoring, Phenology and Irrigation Advisory with Integrated Research Dashboards

TY Minor Project Report

**Submitted in partial fulfillment of the requirements
of the Subject Project Based Learning: Minor Project Lab-I (Sem V)**

by

**Atharva Naik (A-49)
Rahul Meshram (A-47)
Daksh Patel (A-52)**

Supervisor

Dr. Dhanashree Toradmalle



**Department of Computer Engineering
K J Somaiya Institute of Technology
Ayurvihar, Sion, Mumbai 400022
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CERTIFICATE

This is to certify that the project entitled "AgriSense: AI-Driven Crop Monitoring, Phenology and Irrigation Advisory with Integrated Research Dashboards" is the bonafide work of Atharva Naik, Rahul Meshram and Daksh Patel, submitted as a Third Year Semester V Minor Project in Computer Engineering for the subject Project Based Learning: Minor Project Lab-I during the academic year 2026-27.

Dr. Dhanashree Toradmalle

Project Guide

Department of Computer Engineering

Dr. Sarita Ambadekar

Head of Department

Department of Computer Engineering

Dr. Vivek Sunnapwar

Principal

K J Somaiya Institute of Technology, Sion

Place: Sion, Mumbai 400022

Date: _____

PROJECT APPROVAL FOR T. Y.

This project report entitled "AgriSense: AI-Driven Crop Monitoring, Phenology and Irrigation Advisory with Integrated Research Dashboards", submitted by the following students, is presented for approval as a Third Year Minor Project in Computer Engineering.

Atharva Naik (A-49)

Rahul Meshram (A-47)

Daksh Patel (A-52)

EXAMINERS

1. _____

External Examiner Name and Signature

2. _____

Internal Examiner Name and Signature

DECLARATION

We declare that this report presents our project work in our own words. Wherever we have used another person's ideas, methods, observations or datasets, we have acknowledged the original source.

We have distinguished satellite measurements, processed features, published reference observations, derived labels and model predictions throughout the report. The Haridwar field observations belong to their original researchers, and the AgriFieldNet crop labels belong to the credited dataset creators. We do not claim to have collected either dataset.

We have not knowingly fabricated data, experiments or results. Karnal prototype scores are identified as evaluations against derived labels, while the Haridwar and AgriFieldNet results are reported with their respective validation designs and limitations. We understand that the submission remains subject to the Institute's rules on academic integrity.

Atharva Naik: _____

Rahul Meshram: _____

Daksh Patel: _____

Date: _____

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We credit Akash Kumar and Siddhartha Khare for the published Haridwar wheat-stage observations and associated resources. We also acknowledge KVK Dhanauri as the original study site. Radiant Earth Foundation and IDinsight are credited for the AgriFieldNet dataset and its ground-survey crop labels. Our contribution is the additional processing, modelling, evaluation and integration of these sources into AgriSense.

Place: Sion, Mumbai 400022

Date: _____

ABSTRACT

AgriSense brings crop monitoring and remote-sensing research into one web application. Its Karnal dashboard combines satellite observations, weather, soil information, prototype machine-learning predictions and an eight-day irrigation advisory. Two linked research dashboards present wheat phenology results from Haridwar and a multi-crop benchmark built from AgriFieldNet labels. Each component retains its own data source, target definition and validation method.

The Karnal layer contains 43,110 satellite records from 4,311 fixed sample locations over ten dates in the 2025 Kharif season. A 19-period weather calendar provides the application timeline. Optical and radar observations are carried forward independently when a new observation is unavailable, with source dates and observation age retained. Random Forest, an LSTM and XGBoost support crop, phenology and stress inference through a persistent Python service. These targets remain derived labels, so their scores do not establish field-validated agronomic accuracy.

The Haridwar secondary analysis uses published wheat-stage anchors for two seasons and evaluates 59 labelled acquisition rows through leave-one-season-out validation. Its selected fusion Random Forest reaches macro-F1 0.493. The completed AgriFieldNet benchmark covers 5,551 labelled fields, 13 crop classes and 21 ten-day periods. Across four spatially grouped folds, the fusion HistGradientBoosting model obtains 73.30% accuracy and macro-F1 0.391. The gap between these metrics reflects substantial class imbalance.

The application exposes sensor coverage, confusion matrices, model comparisons and optical-gap experiments alongside the operational demonstration. The work establishes an inspectable route from source data to application output, while leaving independent Karnal field validation, irrigation calibration and production access control as further work.

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Karnal interface images are retained from the supplied project report. Research figures are taken from the result files used by the current application. Architecture and UML-style diagrams are redrawn for the complete three-component system.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI / ML	Artificial Intelligence / Machine Learning
API	Application Programming Interface
ARD	Analysis Ready Data
CNN1D	One-dimensional Convolutional Neural Network
ET ₀ / ET _c	Reference / Crop Evapotranspiration
EVI	Enhanced Vegetation Index
FAO	Food and Agriculture Organization
GEE	Google Earth Engine
GIS	Geographic Information System
GRU	Gated Recurrent Unit
K _c	Crop Coefficient
LOSO	Leave-One-Season-Out
LSTM	Long Short-Term Memory
MCC	Matthews Correlation Coefficient
NDMI / NDVI	Normalized Difference Moisture / Vegetation Index
NDRE / NDWI	Normalized Difference Red Edge / Water Index
OOF	Out-of-Fold
RF	Random Forest
SAR	Synthetic Aperture Radar
SAVI	Soil Adjusted Vegetation Index
SD	Standard Deviation
SVM	Support Vector Machine
UI / UML	User Interface / Unified Modeling Language
VV / VH	Co-polarized / Cross-polarized Radar Backscatter
WUA	Water User Association

CHAPTER 1

INTRODUCTION

1.1 Problem Definition

A useful crop-monitoring system must explain both what is happening in a field and how much evidence supports that conclusion. During the monsoon, cloud cover can remove much of the Sentinel-2 record. A clear-looking map can therefore conceal old observations, incomplete sensor coverage or predictions based on uncertain labels. These problems are especially important when the same interface also presents irrigation advice.

AgriSense addresses this problem through three connected components. Karnal demonstrates seasonal monitoring, model inference and water-balance rules. Haridwar examines whether radar and optical features can help distinguish wheat growth stages across seasons. AgriFieldNet provides ground-survey crop labels for a larger spatial classification experiment. These components share a platform, but they do not share a single target or a single accuracy score.

1.2 Aim and Objectives

The aim is to build an understandable, reproducible agricultural intelligence application that connects multisensor data, model evidence and role-specific views.

The objectives are to preserve missing observations and their provenance; align Karnal satellite history with real eight-day weather; serve the selected prototype models; expose transparent irrigation calculations; evaluate Haridwar stage groups using season-held-out testing; evaluate AgriFieldNet crop labels with spatially grouped folds; and publish the results through inspectable dashboards.

The current scope is a working research and decision-support prototype. It does not establish that its irrigation recommendations improve yield or save water, and it does not treat the Karnal sample locations as verified farm boundaries.

INTRODUCTION

1.3 Organization of the Report

Chapter	Purpose
2 Literature	Places sensor fusion, phenology and irrigation methods in context.
3 Requirements	Defines supported user actions, data requirements and runtime needs.
4 Analysis and design	Explains the architecture, role scope, workflows and service relationships.
5 Methodology	Describes the three datasets, preprocessing, validation, robustness tests and APIs.
6 Implementation and results	Presents the interface, data coverage and separately interpreted evaluations.
7 Conclusion and future scope	Summarizes the completed work and the validation still required.
8 References	Lists scientific sources, dataset credits and the project repository.

Reading the evidence

The report uses three distinct kinds of evidence. Karnal coverage statistics describe the observations that were available. Karnal model metrics measure agreement with derived labels. Haridwar and AgriFieldNet results evaluate externally sourced supervision under different holdout schemes. Keeping these categories separate makes the conclusions easier to assess.

The Haridwar targets are five broad stage groups derived from published dates rather than direct field observations at every satellite acquisition. AgriFieldNet supplies observed crop type, but it does not provide field-date phenology or irrigation labels. Its classifier is presented in the research dashboard and has not been substituted for the Karnal inference models.

All application descriptions refer to the source code and result files in the project workspace. The work includes a backend-capable Karnal application and static research pages, which have different runtime requirements.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Introduction

Sentinel-2 contributes spectral information about vegetation, while Sentinel-1 contributes radar backscatter that remains available through cloud [1,2]. Google Earth Engine provides the processing environment for extracting these observations over time [3]. Their combination is well established; the practical challenge in this project is to preserve data quality and explain model results when sources and supervision differ.

2.2 Literature Survey

Table 2.1 Related work	Relevance to AgriSense
Rao et al. [4]; Tufail et al. [5]	Prior multisensor crop classification motivates the optical-radar comparison. Their reported accuracies are not direct benchmarks for this dataset or split.
Wozniak et al. [6]	Temporal radar descriptors motivate seasonal representations rather than a single observation.
Sawant et al. [7]	Field observations provide the context needed to interpret satellite phenology.
Kumar and Khare [8,9]	Published Haridwar wheat-stage dates supply the credited anchors for the secondary five-class analysis.
Radiant Earth and IDinsight [10]; Tuo et al. [11]	AgriFieldNet is an existing crop benchmark with prior competition solutions.
Allen et al. [16]	Reference evapotranspiration and crop coefficients provide the methodological basis for the Karnal water-demand calculation.

The contribution of AgriSense lies in the implementation and evaluation built around these sources. The application connects data preparation, model inspection and stakeholder views, while retaining the differences between operational demonstrations and research experiments.

REVIEW OF LITERATURE

2.3 Limitations of Existing Systems

Integration introduces problems that a single classifier score cannot resolve. Observation frequency differs across satellites and weather products. The same vegetation signal can mean different things for different crops or growth stages. Models may also reproduce a labelling rule without learning a relationship that holds in the field.

Table 2.2 Integration gap	Project response and remaining limitation
Cloud-dependent optical input	Retain radar and audit optical availability; radar coverage is still checked rather than assumed complete everywhere.
Unclear label origin	Separate published stage anchors, surveyed crop labels and Karnal pseudo labels.
Mismatched calendars	Use an eight-day Karnal display calendar while retaining source dates; do not invent intermediate satellite observations.
Class imbalance	Show macro-F1, per-class support and confusion matrices alongside accuracy.
Overstated generalization	Use LOSO and spatial groups, while acknowledging the absence of a separate final test after model selection.
Irrigation without outcome data	Expose the formula and thresholds; require local calibration and observed outcomes before operational claims.
Interface roles mistaken for security	Treat current roles as presentation rules; server-side authentication and authorization remain further work.

The review therefore supports a bounded research question: how well can one application preserve the meaning and limitations of several agricultural data products? It does not justify claiming the first optical-radar crop classifier, the first Haridwar phenology analysis or a validated automated irrigation service.

CHAPTER 3

REQUIREMENT SPECIFICATION

3.1 Introduction

AgriSense must support both seasonal exploration and careful inspection of model evidence. Its requirements reflect the application that is implemented: a Karnal operational demonstration, a Haridwar research page and an AgriFieldNet benchmark page.

3.2 Functional Requirements

The Karnal user can select a role, choose a master period, play the seasonal timeline, switch among permitted map layers, inspect a sample location and export its data. The field view presents observation provenance, weather, water-balance output and, for supported roles, predictions from the persistent inference service.

The five implemented personas are Agriculture Officer, Irrigation Officer, Water User Association, Farmer and Policy Maker. The farmer view shows two assigned demonstration plots. Irrigation and WUA views emphasize deficit and priority. The policy view summarizes the region. These personas control what the interface displays; they are not authenticated accounts.

The Haridwar page must expose raw and smoothed NDVI, stage anchors, sensor-family results, confusion matrices, transition estimates and optical-gap experiments. The AgriFieldNet page must support searchable model results, feature-set filters, per-fold and per-crop inspection, count or row-percentage confusion matrices, coverage, importance, figures and CSV downloads.

3.3 Non-Functional Requirements

Outputs must identify their source and validation status. Missing data must remain distinguishable from low vegetation values. A dashboard build must reject inconsistent AgriFieldNet predictions or scores. The browser should report unavailable services rather than inventing predictions. Figures, tables and labels must remain readable, and research outputs should work on a static host without model-serving dependencies.

REQUIREMENT SPECIFICATION

3.4 Hardware Requirements

Table 3.1 Component	Development guidance
CPU and memory	A multicore workstation and sufficient RAM for raster processing and model fitting; 16 GB is a practical development target, not a measured minimum.
Storage	Space for raster exports, feature tables, notebooks, serialized models and result images; requirements depend on retained exports.
Training accelerator	GPU support is optional for the application and useful for supported training libraries. The saved benchmark used a local Jupyter environment.
Client	A current desktop browser; internet access is needed for external map tiles and externally loaded frontend assets.

3.5 Software Requirements

Table 3.2 Layer	Technology used
Geospatial processing	Google Earth Engine; Python, Rasterio, PyProj and GeoPandas where required.
Data preparation	Pandas, NumPy and standard-library CSV/JSON processing.
Classical models	scikit-learn, XGBoost, CatBoost and LightGBM in the research workflow.
Temporal models	TensorFlow/Keras for LSTM, GRU and CNN1D experiments.
Application server	Node.js and Express; FastAPI/Uvicorn for persistent inference.
Browser interface	HTML, CSS and JavaScript; Leaflet and Chart.js in the Karnal interface.
Research publication	Local JSON snapshots, CSVs and PNG figures; the AgriFieldNet page adds no external chart service.
Version control	Git and the AgriSense repository [12].

CHAPTER 4

PROJECT ANALYSIS AND DESIGN

4.1 Introduction

The application separates data preparation from display. Large geospatial operations and model evaluation happen offline; the browser reads prepared files and calls the inference service only when required. This keeps a model-training experiment from becoming an implicit dependency of every dashboard visit.

4.2 System Architecture

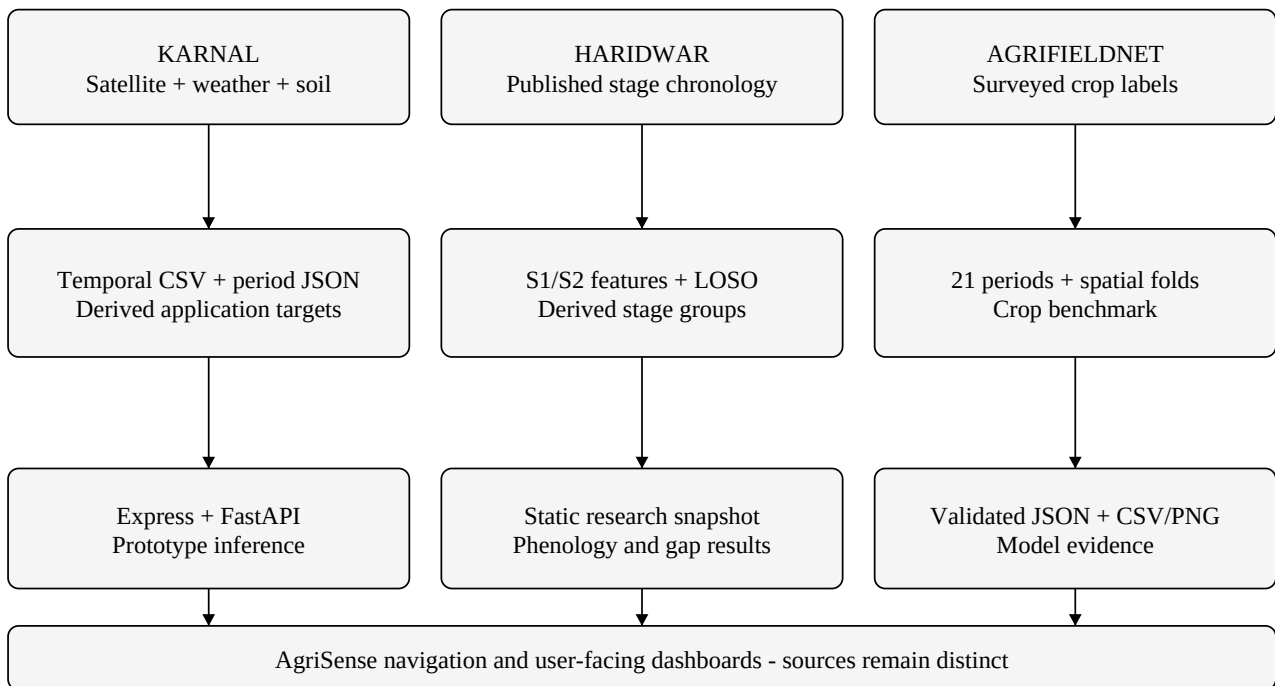


Figure 4.1: System architecture across Karnal, Haridwar and AgriFieldNet.

Karnal uses a satellite history, weather and soil products to build nineteen period files. Express serves these files and proxies selected field requests to FastAPI, whose model registry loads the saved artifacts once. Haridwar and AgriFieldNet instead use static result snapshots and figures. Navigation connects the three pages without merging their labels or claiming that all research models are deployed.

The current server blocks the `private_models` path from static access. This protects that route but does not provide a full production security design. Deployment also needs an explicit public-file boundary, authenticated user scopes and protection for operations such as rebuilding data.

PROJECT ANALYSIS AND DESIGN

4.3 System Flow Diagram

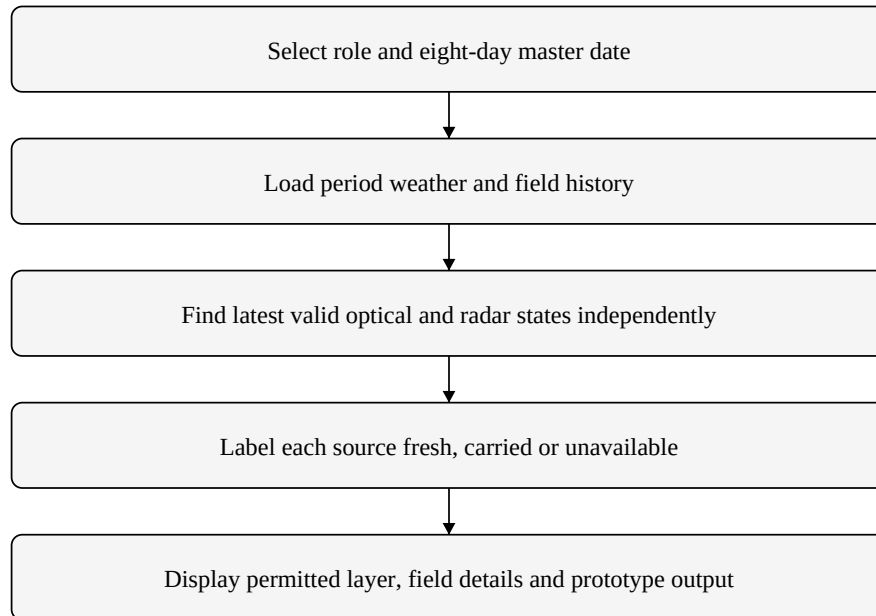


Figure 4.2: Activity flow for selecting observations and presenting a period.

4.4 UML Diagrams

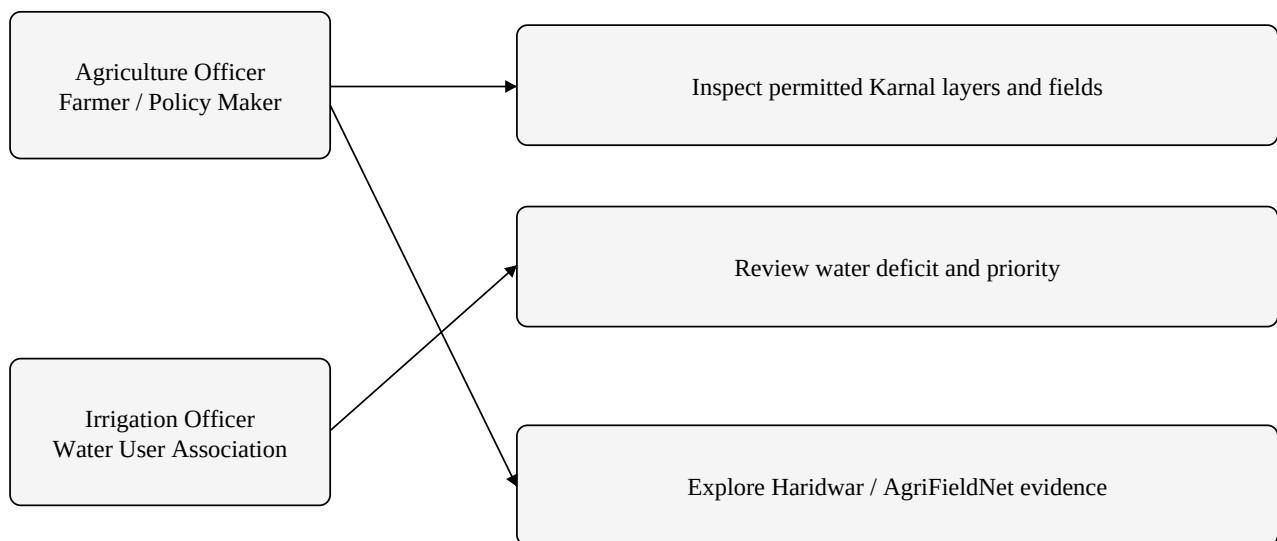


Figure 4.3: Use cases and the distinction between presentation roles and services.

Role selection controls the visible tasks. Data preparation and model fitting are separate maintenance activities. A user can inspect the research pages without starting the inference service, while live Karnal model output depends on the Python runtime and its artifacts.

PROJECT ANALYSIS AND DESIGN

4.4.1 Sequence Diagram

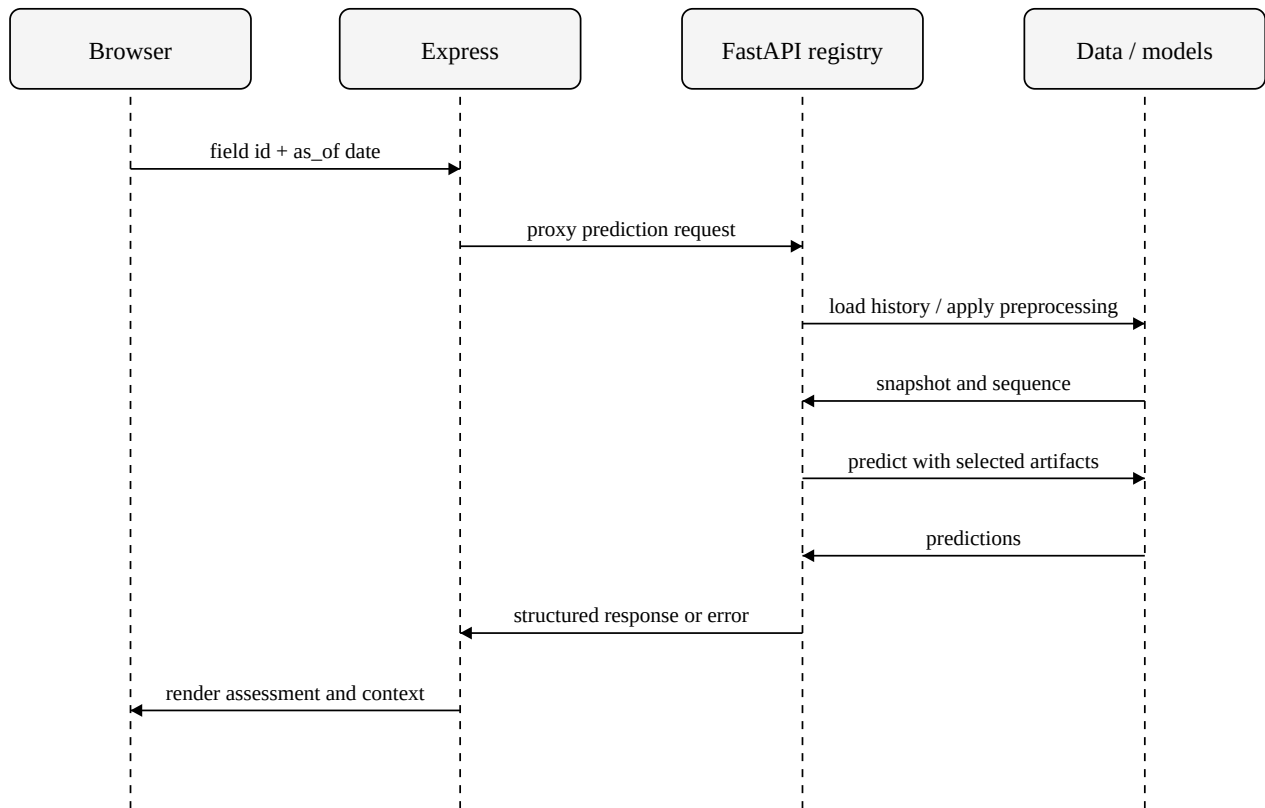


Figure 4.4: Field inference request and response sequence.

After a user selects a field and date, the browser requests `/api/inference/field/{fieldId}` with an `as_of` value. Express forwards the request to the Python service. The registry looks up the field history, constructs the relevant snapshot and temporal input, applies the saved preprocessing objects and invokes the selected models.

The service returns predictions and model context to the browser. If the service or required artifact is unavailable, the API returns an error state. Stored dashboard values and live inference are therefore separate paths; a static page cannot run the Python models merely because it can display earlier results.

The research publication path is simpler: a browser fetches its JSON snapshot, then renders tables and figures. For AgriFieldNet, a build-time reconciliation step checks the relationship between metrics, class support and out-of-fold predictions before a new snapshot is written.

PROJECT ANALYSIS AND DESIGN

4.4.2 Class and Data Relationship Diagram

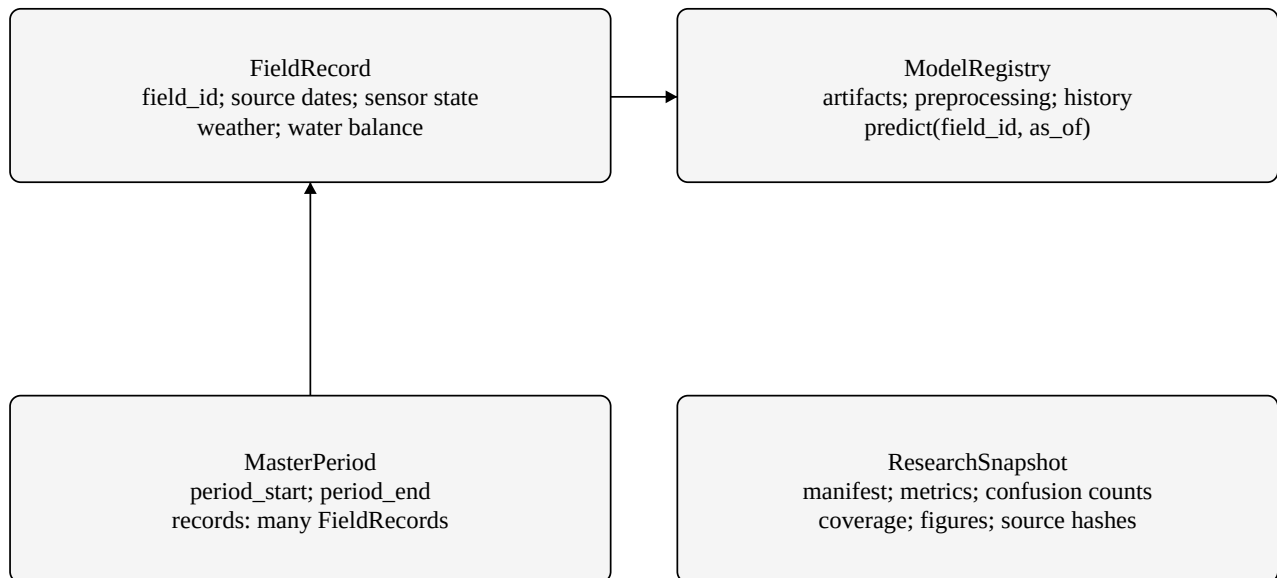


Figure 4.5: Logical data objects and services; this is a design view, not an ORM schema.

4.5 Project Timeline

Table 4.1 Phase	Current outcome
Data foundation	Karnal satellite history, weather periods and soil context prepared.
Operational prototype	Temporal display, five personas and persistent inference implemented.
Haridwar integration	Two-season phenology results and robustness outputs connected.
AgriFieldNet benchmark	Eight feature batches and eight observation batches consolidated; four-fold evaluation completed.
Research publication	Metrics reconciled, dashboard interactions added and source downloads provided.
Further validation	Independent Karnal labels, irrigation calibration and production controls remain future work.

The phases describe the development sequence rather than verified calendar durations. Training and dashboard integration are complete for the saved research outputs; further agronomic validation is a separate activity.

CHAPTER 5

METHODOLOGY

5.1 Introduction

The methodology preserves each dataset's observation unit and target. Karnal is indexed by sample location and date. Haridwar uses acquisitions within two wheat seasons at one experimental site. AgriFieldNet uses a labelled field as the prediction unit and a temporal sequence as its input.

5.2 Proposed Methodology

Table 5.1 Component	Data and supervision
Karnal	4,311 sample locations; 10 satellite dates; 43,110 rows; 19 weather periods; 24 soil bands. Crop, stage and stress targets are derived.
Haridwar	68 calendar rows, 59 labelled acquisitions, 20 published stage anchors and 101 S1 track observations across two seasons [8,9].
AgriFieldNet	5,551 surveyed training fields, 13 classes and 116,571 field-period rows in 21 ten-day bins [10]. No observed phenology target.

5.2.1 Karnal Data and Temporal Alignment

The local satellite pipeline supports both fourteen-band optical-plus-radar tiles and three-band radar-only tiles. It samples the same locations across ten dates. The model feature `VV_VH_ratio_dB` is computed as `VV` minus `VH` because the source backscatter is in decibels. Missing optical values remain missing in the observation table.

The eight-day master calendar joins fresh weather with the latest valid optical and radar observations available on or before the selected date. Each modality retains its own source date and age. A carried optical value is usable context, but it is not a new clear-sky measurement. The 250 m soil stack provides regional context and does not become a field-scale measurement when sampled at a point.

METHODOLOGY

5.2.2 Karnal Models and Irrigation Rules

Table 5.2 Engine	Selected model	Target and input
Crop	Random Forest	Paddy versus Unclassified; optical-radar snapshot.
Phenology	LSTM	Five derived NDVI stage classes; three-step sequence.
Stress	XGBoost	Derived moisture-stress classes; optical-radar snapshot.
Irrigation	Rules	Period weather, crop coefficient and moisture context.

The LSTM uses vegetation and moisture indices, VV, VH, their dB difference, validity flags and season day. Field groups are split into 3,017 training, 647 validation and 647 test locations. This prevents the same location from appearing in multiple splits, but it does not turn the NDVI-derived target into independently observed phenology. The snapshot phenology baselines and temporal LSTM also have different class sets and evaluation units, so their raw accuracies are not a like-for-like ranking.

The water-demand calculation estimates FAO-56-style ET0 from meteorological inputs and computes $ET_c = K_c \times ET_0$ [16]. The implementation then uses $P_{eff} = 0.80 \times \text{rainfall}$ and deficit = $\max(0, ET_c - P_{eff})$. These last two rules are prototype simplifications. ERA5 potential evaporation is kept distinct from the estimated ET0.

Deficit bands below 5, 15 and 30 mm, and at least 30 mm, set increasing priority and depth rules. Recommended depth is scaled by 1.15 and capped by band. Very low moisture can raise the numeric priority and shorten the displayed timing without recomputing the base depth or label. Soil properties are displayed as context; the rule is not a complete root-zone storage, runoff, percolation or irrigation-efficiency model.

METHODOLOGY

5.2.3 Haridwar Phenology Analysis

Akash Kumar and Siddhartha Khare collected and published the source wheat-stage observations at KVK Dhanauri, Haridwar [8,9]. AgriSense uses the published chronology for 2023-24 and 2024-25 as external evidence. It does not claim the original field campaign, PhenoCam observations or published date-estimation results.

Ten stages per season are grouped into Establishment, Vegetative, Reproductive, Grain Filling and Senescence/Harvest. Acquisition labels are derived using intervals whose boundaries fall between adjacent stage anchors. The resulting 59-row table contains 12 Establishment, 23 Vegetative, 16 Reproductive, seven Grain Filling and one Senescence/Harvest row.

S2-only, S1-only and fused features are evaluated by leaving one entire season out at a time. Preprocessing and model fitting use the training season, and predictions from the two held-out seasons are pooled. Macro-F1 is emphasized because accuracy can hide poor performance on the rare stages. These rows describe one site repeatedly over time; they are not 59 independent farms.

The dashboard also compares raw and Whittaker-smoothed NDVI and presents transition estimates from Whittaker and Double Logistic methods. The saved transition summary uses interval error in days. It is separate from stage-classification macro-F1 and from the original paper's validation of phenological dates.

Additional optical masking tests selected models under missing S2 information while retaining radar in the fusion input. This controlled experiment is distinct from the observed natural cloud gaps. Model selection uses the same LOSO results that are reported, so the selected scores are comparisons within this study rather than an untouched final-test estimate.

METHODOLOGY

5.2.4 AgriFieldNet Spatial Benchmark

AgriFieldNet crop labels are credited to Radiant Earth Foundation and IDinsight [10]. The project reconstructs labelled training fields and attaches 21 ten-day feature bins. The saved feature batches run from 1 November 2021 to 29 May 2022. This temporal extension is an AgriSense-derived product, not the original single-composite challenge input.

Table 5.3 Family	Representation
A0	Static Sentinel-2 summaries.
A1	Sentinel-2 temporal features.
A2	Sentinel-1 temporal features.
A3	Combined Sentinel-1 and Sentinel-2 temporal features.
A4	Compact temporal descriptors.

The completed run compares sixteen classical models in each of five families, plus CNN1D, GRU and LSTM sequence models. Representative field points define 0.10-degree spatial grid groups. Four stratified group folds hold groups apart, and every training fold must contain all classes. Median imputation, missing-value indicators and scaling are fitted on the training partition before transformation of held-out data.

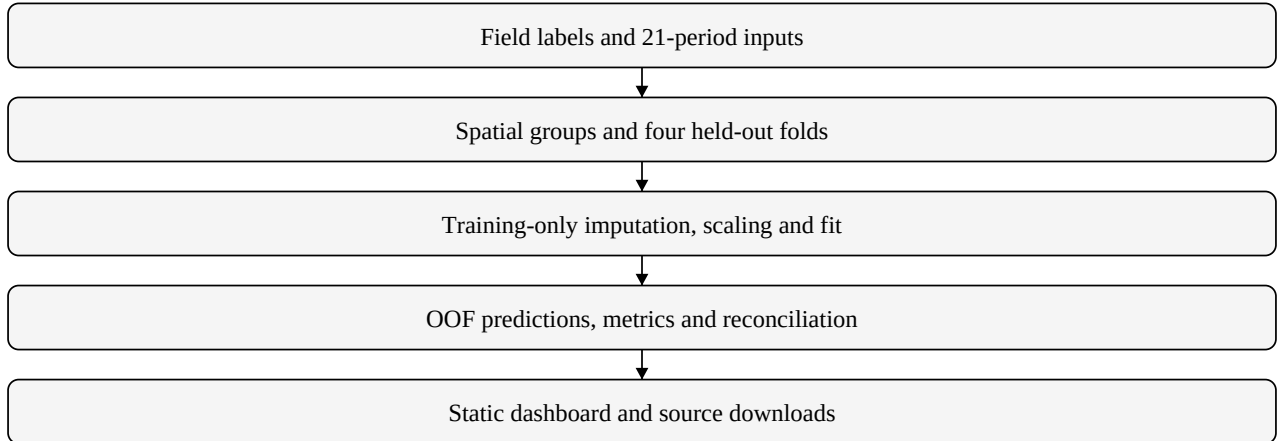


Figure 5.1: Training, held-out prediction and dashboard publication.

Scores are computed from pooled out-of-fold predictions. The selected leader is chosen from these same results; spatial grouping reduces local overlap but is not a separate regional-transfer or nested model-selection test.

METHODOLOGY

5.2.5 Optical Gaps and Feature Importance

The AgriFieldNet robustness experiment fits the selected optical and fusion models once per fold, then perturbs held-out optical inputs. Random scenarios mask approximately 20%, 40% or 60% of the 21 periods for each field. Consecutive scenarios remove two or four ten-day bins. The baseline is evaluated once; each non-baseline scenario has fifteen masking repeats.

The masking code replaces S2 signal columns with missing values. It leaves radar inputs and optical quality/count diagnostics in place. As a result, the experiment measures sensitivity to missing optical signal values under this particular input design; it is not a complete physical simulation of clouds. The repeat standard deviation describes variation across masks, not uncertainty in performance on a new region.

Natural coverage is audited separately using NDVI validity, clear-observation fractions and radar availability. AgriFieldNet radar is not present in every field-period, unlike the complete S1 coverage in the saved Karnal table. The distinction prevents a dataset-specific coverage result from becoming a general claim about all radar observations.

Exploratory Permutation Importance

The saved fusion importance uses the quick configuration: two permutations per feature and up to 400 held-out fields per fold. A fresh fold-local pipeline is fitted, then each original feature is permuted and the change in macro-F1 is recorded. Signed values are retained, including negative values.

The dashboard ranks individual features by the magnitude of their mean signed contribution. The original saved figure instead sums positive contributions by base feature. These are different summaries. Correlated spectral and temporal features can share importance, and this small evaluation sample may underrepresent rare crops. The estimates help explore model behavior but do not establish causal drivers or full-budget importance.

METHODOLOGY

5.3 Data and API Schema

Table 5.4 Route or file	Purpose
GET /api/data	Master metadata and map data.
GET /api/period/{date}	Records for one eight-day master period.
GET /api/field-timeline/{id}	History for a selected sample location.
GET /api/model-evaluation	Stored prototype scores and confusion matrices.
GET /api/model-metadata	Stored model descriptions and metadata.
GET /api/inference/health	Inference registry readiness.
GET /api/inference/models	Selected models and availability.
GET /api/inference/field/{id}?as_of=date	Prediction request for a field and selected date.
POST /api/run-pipeline	Local data rebuild; not a browser-side training job.
data/haridwar/haridwar_dashboard.json	Static Haridwar evidence and figure references.
data/agrifieldnet/dashboard.json	Validated benchmark snapshot, support, coverage and provenance.

The Karnal master file describes nineteen periods, while period JSON files contain the observation state, weather and water-balance output for each location. The real temporal CSV remains the source for model sequence history. Serialized model and preprocessing artifacts are stored separately in `private_models`.

The AgriFieldNet builder reads saved outputs, verifies complete and unique held-out field coverage, and recalculates metrics from 460,733 predictions across 83 model configurations. It publishes aggregate confusion counts, compact CSVs and source hashes. A new snapshot is rejected when source results disagree.

CHAPTER 6

PROJECT IMPLEMENTATION AND RESULTS

6.1 Introduction

AgriSense combines an operational prototype with two research dashboards. This chapter reports their outputs separately so that a reader can distinguish successful software integration from independent agronomic validation.

6.2 System Implementation

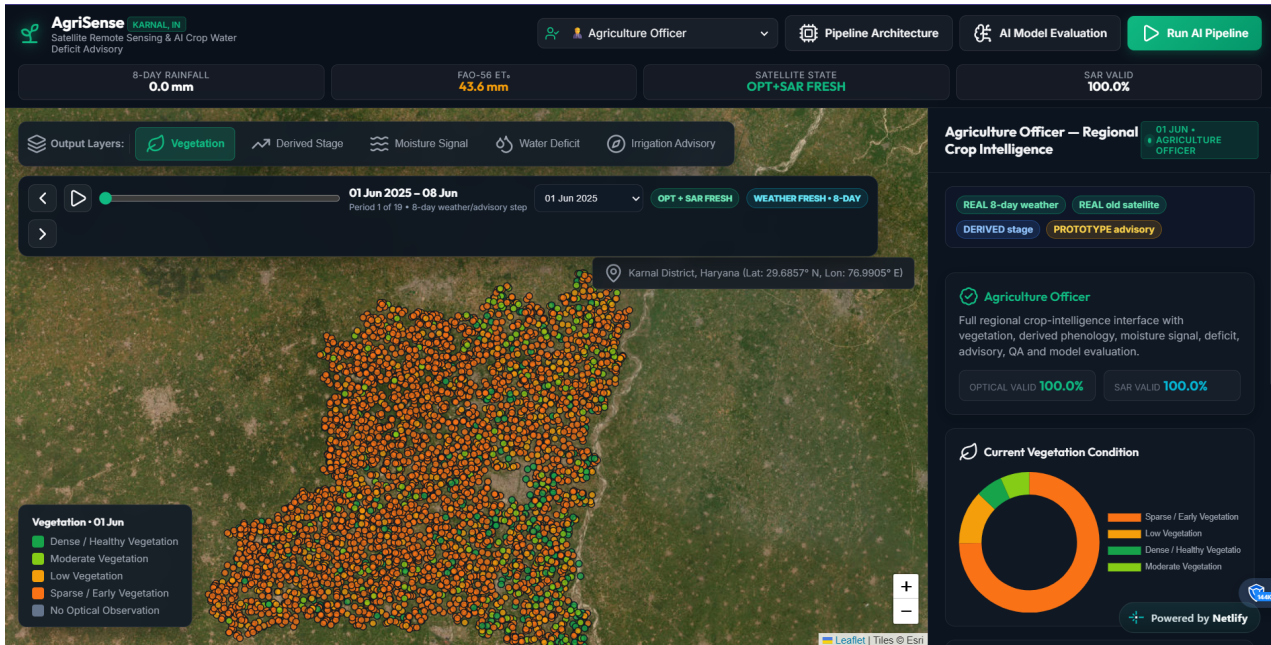


Figure 6.1: Karnal agriculture officer interface retained from the supplied report.

The Karnal interface uses Leaflet for mapping and Chart.js for analytics. Its period selector, slider and playback controls traverse the nineteen master steps. A field inspector brings together source age, vegetation and moisture context, rainfall, ET0, deficit and prototype advice. Five personas tailor the visible controls and summaries.

Express is the browser-facing server, normally on port 3000. It can start a persistent FastAPI service on port 8001. The registry reuses loaded models rather than reopening artifacts for every request. The Karnal data views can fall back to prepared static files, while live inference still requires the backend. Haridwar and AgriFieldNet are linked directly from the application headers.

PROJECT IMPLEMENTATION AND RESULTS

6.2.1 Field Inspection and Research Pages

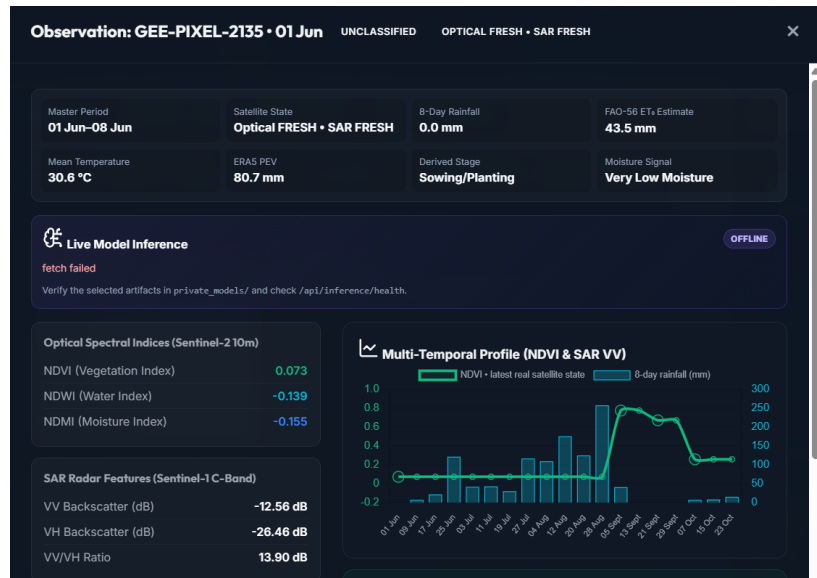


Figure 6.2: Karnal field inspector retained from the supplied report.

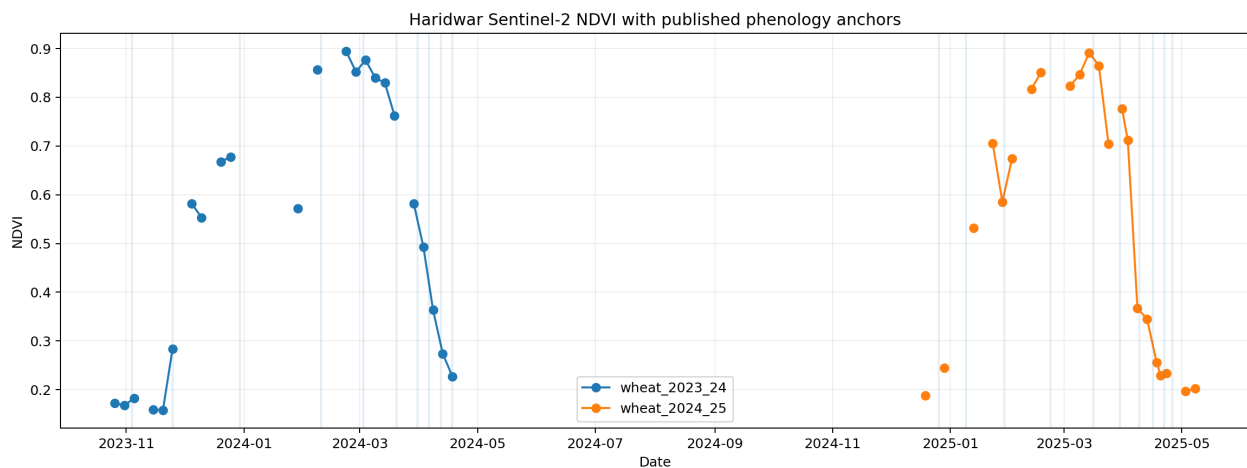


Figure 6.3: Haridwar NDVI and published stage anchors in the saved research outputs.

The Haridwar page provides an overview, phenology exploration, model analysis, cloud robustness and the original figure gallery. It contains no Haridwar weather-driven irrigation output. The AgriFieldNet page exposes all 83 model configurations, sortable results, per-crop and per-fold metrics, confusion matrices, class support, natural coverage, synthetic masking tests and source downloads.

The research pages publish evaluation evidence. They do not add an AgriFieldNet inference endpoint or a live map of newly classified farms. This separation lets users inspect completed research without mistaking it for a deployed advisory model.

PROJECT IMPLEMENTATION AND RESULTS

6.3 Performance Evaluation

6.3.1 Karnal Coverage and Prototype Models

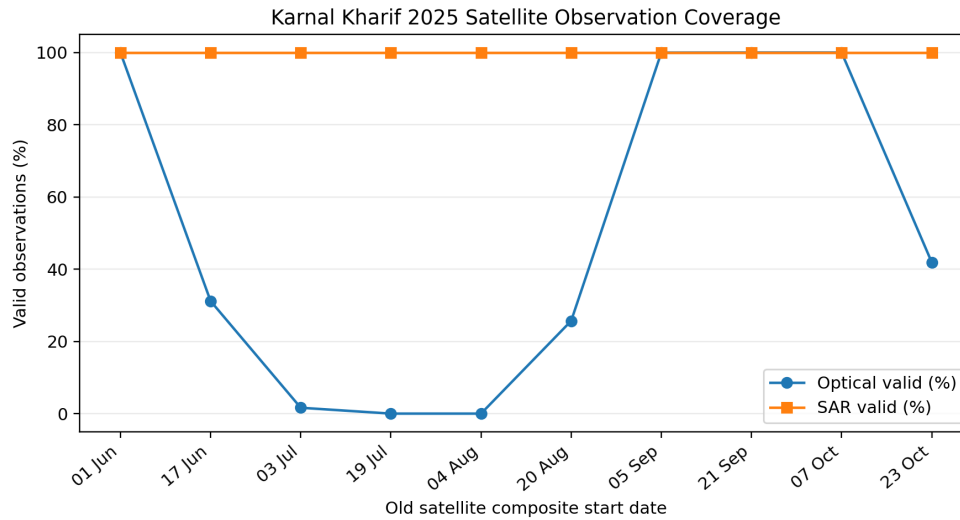


Figure 6.4: Natural optical and radar coverage in the ten-date Karnal table.

The ten-date table contains 4,311 locations per date. Optical validity falls from 100% on 1 June to 31.18% on 17 June, 1.62% on 3 July and 0% on both 19 July and 4 August. Radar validity remains 100% for these saved dates. This is an observation-availability result, not a crop-accuracy measurement.

Table 6.1 Stored prototype model	Accuracy	Macro-F1
Crop / Random Forest	100.00%	1.000
Crop / XGBoost	100.00%	1.000
Crop / SVM RBF	99.65%	0.981
Phenology / Random Forest	100.00%	1.000
Phenology / XGBoost	99.77%	0.997
Phenology / LSTM	80.89%	0.740
Stress / Logistic Regression	98.96%	0.733
Stress / Random Forest	99.77%	0.874
Stress / XGBoost	100.00%	1.000

These scores measure agreement with derived targets. Crop and stress perfect scores do not demonstrate independently validated crop identity or moisture stress. The LSTM uses five pseudo-stage classes and a field-group temporal test; its snapshot baselines use three classes. The nineteen weather periods support changing water-demand estimates, but no irrigation outcome accuracy has been measured.

PROJECT IMPLEMENTATION AND RESULTS

6.3.2 Haridwar Phenology Results

Table 6.2 Feature family / model	Accuracy	Macro-F1
S2 only / RBF SVM	55.93%	0.461
S1 only / Random Forest	64.41%	0.429
S1 + S2 / Random Forest	64.41%	0.493

Fusion has the highest macro-F1 among the selected sensor-family leaders. The S1-only model has the same overall accuracy as fusion but a lower macro-F1, showing that correct predictions are distributed differently across stages. The five-class table is small and includes only one Senescence/Harvest observation, so these results remain sensitive to class support and season differences.

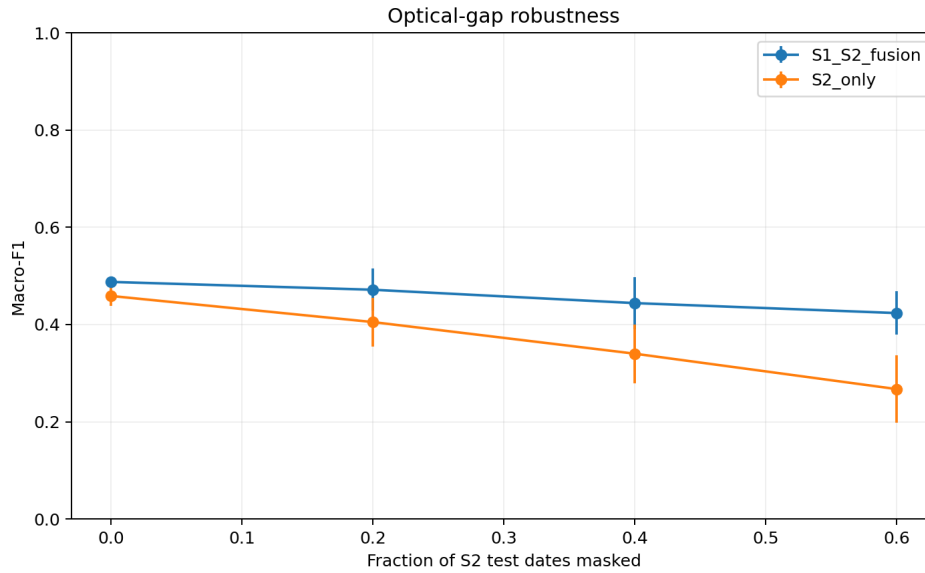


Figure 6.5: Additional optical masking in the Haridwar secondary analysis.

At 60% additional optical masking, the fusion model retains mean macro-F1 about 0.424, compared with 0.267 for the optical-only leader. Natural optical usability is 60.5% in 2023-24 and 73.3% in 2024-25, with approximate longest missing runs of 30 and 10 days. These natural gaps and the masking experiment answer different questions.

The saved transition summary reports mean interval error of 8.50 days for Double Logistic and 8.67 days for Whittaker. These secondary estimates must not be relabelled as the original source paper's date-validation results.

PROJECT IMPLEMENTATION AND RESULTS

6.3.3 AgriFieldNet Classification Results

Table 6.3 Family / selected model	Accuracy	Macro-F1
A0 static S2 / CatBoost	57.74%	0.260
A1 temporal S2 / CatBoost	69.48%	0.389
A2 temporal S1 / CatBoost	68.85%	0.340
A3 fusion / HistGradientBoosting	73.30%	0.391
A4 descriptors / CatBoost	70.71%	0.371

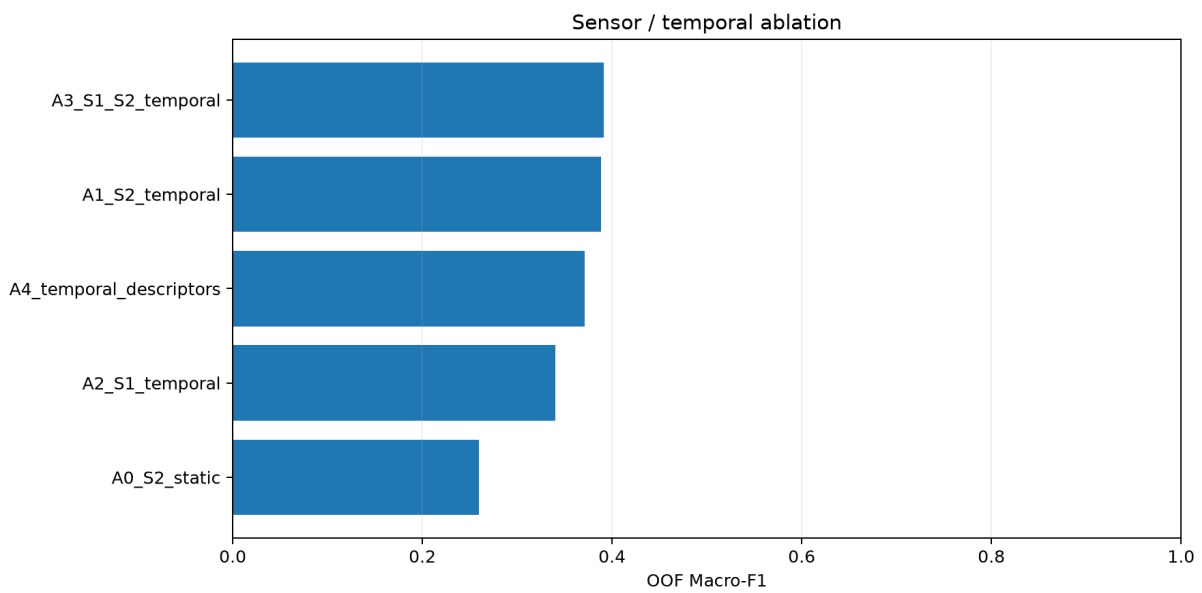


Figure 6.6: Selected sensor-family leaders under spatially grouped evaluation.

The completed benchmark contains 116,571 rows from 5,551 fields and 21 periods. All eight periodic batches and eight observation batches are present. Each of the 83 configurations has held-out predictions covering the complete field set; the dashboard stores the reconciled results.

Fusion reaches pooled OOF macro-F1 0.390976 and accuracy 0.733021. Its balanced accuracy is 0.379838 and weighted F1 is 0.717539. The temporal optical leader is close at macro-F1 0.388634, a difference of only 0.002342. Since leaders use different model families and no independent significance test is reported, this difference does not establish a general fusion advantage.

PROJECT IMPLEMENTATION AND RESULTS

6.3.4 Rare Classes and Deep Models

Table 6.4 Crop	Fields	Fusion F1
Bersem	16	0.190
Coriander	14	0.000
Garlic	48	0.513
Gram	59	0.679
Green pea	23	0.000
Lentil	103	0.015
Maize	293	0.760
Mustard	990	0.649
No crop/Fallow	1641	0.778
Potato	41	0.078
Rice	129	0.132
Sugarcane	163	0.464
Wheat	2031	0.825

Wheat, No crop/Fallow and Mustard account for most fields. The fusion leader has zero F1 for Coriander and Green pea, and very low F1 for Lentil, Potato and Rice. Its 73.30% accuracy must therefore be read alongside the much lower macro-F1. High aggregate accuracy is not uniform crop recognition.

Table 6.5 Deep model	OOF accuracy	OOF macro-F1
CNN1D	57.45%	0.343
GRU	50.37%	0.318
LSTM	51.56%	0.319

None of the three sequence models exceeds the selected classical fusion leader in pooled macro-F1 in this run. This is a result for the saved architectures and training settings, not a claim that recurrent or convolutional models are inherently unsuitable for crop classification. The research LSTM is also a different model and task from the Karnal pseudo-phenology LSTM.

PROJECT IMPLEMENTATION AND RESULTS

6.3.5 Robustness and Interpretation

Table 6.6 Masking scenario	S2 macro-F1	Fusion macro-F1
Baseline	0.389	0.391
20% random	0.363 +/- 0.006	0.367 +/- 0.010
40% random	0.306 +/- 0.008	0.327 +/- 0.008
60% random	0.194 +/- 0.004	0.259 +/- 0.007
20-day consecutive	0.375 +/- 0.009	0.377 +/- 0.006
40-day consecutive	0.350 +/- 0.008	0.362 +/- 0.007

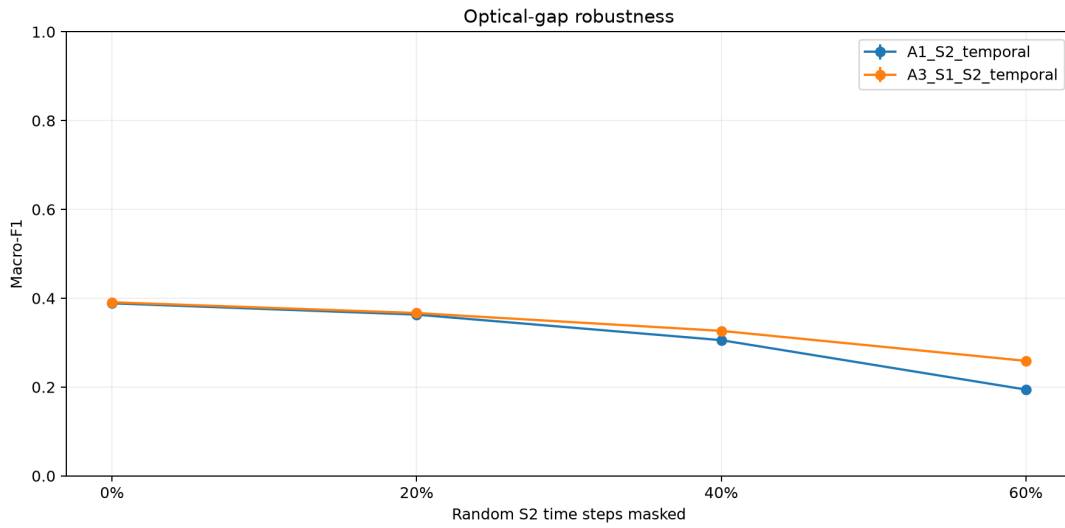


Figure 6.7: AgriFieldNet random optical-gap experiment; bars show repeat SD.

Under 60% random masking, fusion loses about 0.132 macro-F1 from baseline, while optical-only loses about 0.194. This supports greater resilience for the selected fusion pipeline under the tested perturbation. The values after masking are still modest, and the test retains optical quality diagnostics. The fifteen-repeat standard deviations are not confidence intervals for geographic generalization.

The application makes these qualifications visible through per-class results, model details and coverage views. The saved quick permutation estimates are exploratory. Overall, the completed evidence supports multisensor integration and inspectable evaluation; it does not support claims of universally accurate crop mapping, field-validated Karnal phenology or proven irrigation savings.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

AgriSense now connects three complete application components. Karnal demonstrates a seasonal workflow from real satellite and weather data to map inspection, persistent model inference and transparent advisory rules. Haridwar adds a credited two-season phenology analysis. AgriFieldNet adds a completed multi-crop spatial benchmark and a dashboard that exposes the results beyond a single headline score.

The main engineering result is the continuity of the evidence trail. A user can inspect the age of an observation, the source of a target, the performance of a model and the effect of missing optical data. The benchmark builder also verifies that published metrics agree with held-out predictions before refreshing the AgriFieldNet snapshot.

The quantitative results are mixed in a useful way. Haridwar fusion attains macro-F1 0.493, while AgriFieldNet fusion attains 0.391 and retains more performance than its selected optical-only comparison under severe masking. Rare-class errors remain substantial. These findings give the next development cycle a clearer direction than accuracy alone would provide.

7.2 Limitations

Karnal crop, stage and stress targets are derived; its model scores are not independent agronomic validation. Haridwar covers two seasons at one site, with acquisition labels inferred from stage anchors and only one terminal-stage row. AgriFieldNet leaders are chosen using the reported OOF results, and spatial grid grouping does not establish cross-region transfer.

The water rule does not model the full root-zone balance or demonstrate measured water savings. Coarse weather and soil inputs do not gain field-scale detail through resampling. Current personas provide interface filtering rather than secure access control. Static research outputs are not deployed real-time research-model inference.

CONCLUSION AND FUTURE SCOPE

7.3 Future Scope

Independent Agronomic Validation

The first priority is to collect independent crop, phenology and moisture observations in Karnal, with clear field identifiers and dates. These measurements should replace derived targets and support prospective testing on later seasons. Verified field boundaries would also improve the interpretation of point-based outputs.

Research Evaluation

Further work should reserve a final spatial or temporal holdout after model selection, assess uncertainty using the appropriate independent unit, and investigate rare-class performance. The Haridwar analysis needs more sites, seasons and observations near senescence. The AgriFieldNet temporal extraction window and source hashes should be included in a frozen release, with full-budget feature importance clearly separated from the quick exploratory run.

Robustness experiments can be extended by masking quality flags together with optical signals, testing realistic cloud patterns and comparing matched model families. These experiments would help distinguish the effect of sensor information from the effect of choosing a different estimator.

Irrigation and Application Deployment

Local agronomic calibration should cover crop coefficients, effective rainfall, irrigation efficiency and priority thresholds. Soil-moisture measurements, irrigation logs and water-availability data are needed before claiming practical advisory benefit. The current calculation can then be extended to a stateful root-zone water balance.

Production work should add authenticated users, server-side field scopes, protected administrative actions and explicit public asset directories. Usability testing should include farmers and officers rather than relying only on developer inspection. Multilingual and mobile access can follow once the information and advisory content are validated.

These steps extend the completed prototype without treating research-dashboard integration as proof of operational readiness.

CHAPTER 8

REFERENCES

- [1] R. Torres et al., "GMES Sentinel-1 mission," *Remote Sensing of Environment*, 120, 9-24, 2012. doi:10.1016/j.rse.2011.05.028.
- [2] M. Drusch et al., "Sentinel-2: ESA's Optical High-Resolution Mission for GMES Operational Services," *Remote Sensing of Environment*, 120, 25-36, 2012. doi:10.1016/j.rse.2011.11.026.
- [3] N. Gorelick et al., "Google Earth Engine: Planetary-scale geospatial analysis for everyone," *Remote Sensing of Environment*, 202, 18-27, 2017. doi:10.1016/j.rse.2017.06.031.
- [4] P. Rao et al., "Using Sentinel-1, Sentinel-2, and Planet Imagery to Map Crop Type of Smallholder Farms," *Remote Sensing*, 13(10), 1870, 2021. doi:10.3390/rs13101870.
- [5] R. Tufail et al., "A machine learning approach for accurate crop type mapping using combined SAR and optical time series data," *Advances in Space Research*, 69, 331-346, 2022. doi:10.1016/j.asr.2021.09.019.
- [6] E. Wozniak et al., "Multi-temporal phenological indices derived from time series Sentinel-1 images to country-wide crop classification," *International Journal of Applied Earth Observation and Geoinformation*, 107, 102683, 2022. doi:10.1016/j.jag.2022.102683.
- [7] S. A. Sawant, J. D. Mohite and S. Pappula, "Integration of human participatory sensing and archives of remote sensing observations for field level crop phenology estimation," *ISPRS Archives*, XLII-4, 547-550, 2018. doi:10.5194/isprs-archives-XLII-4-547-2018.
- [8] A. Kumar and S. Khare, "Evaluation of PlanetScope and Sentinel-2 derived wheat crop phenology using infrared-enabled PhenoCam and field observations," *Ecological Informatics*, 96, 103821, 2026. doi:10.1016/j.ecoinf.2026.103821.
- [9] A. Kumar and S. Khare, associated data and analysis resources, Figshare, version 2, 2026. doi:10.6084/m9.figshare.30162856.v2.
- [10] Radiant Earth Foundation and IDinsight, "AgriFieldNet Competition Dataset," version 1.0, 2022. doi:10.34911/rdnt.wu92p1.
- [11] M. Tuo, C. Emelike and T. Ogundare, "AgriFieldNet Model for Crop Detection from Satellite Imagery," version 1.0, first-place challenge solution. doi:10.34911/rdnt.k2ft4a.
- [12] AgriSense Project Team, "AgriSense," project source code and research outputs, 2026. <https://github.com/Atharva-cyber849/AgriSense>. Local implementation and result snapshot reviewed 25 September 2026.
- [13] C. Funk et al., "The climate hazards infrared precipitation with stations - a new environmental record for monitoring extremes," *Scientific Data*, 2, 150066, 2015. doi:10.1038/sdata.2015.66.
- [14] J. Munoz-Sabater et al., "ERA5-Land: a state-of-the-art global reanalysis dataset for land applications," *Earth System Science Data*, 13, 4349-4383, 2021. doi:10.5194/essd-13-4349-2021.
- [15] OpenLandMap, soil property layers at standard depths, Google Earth Engine Data Catalog. <https://developers.google.com/earth-engine/datasets/catalog/OpenLandMap>.
- [16] R. G. Allen, L. S. Pereira, D. Raes and M. Smith, *Crop Evapotranspiration: Guidelines for Computing Crop Water Requirements*, FAO Irrigation and Drainage Paper 56, Rome, 1998. <https://www.fao.org/4/x0490e/x0490e00.htm>.